Tapon Kumer Ray

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Research Interests: Human-Computer Interaction, AR Interfaces, User-Centered Design, Intelligent Agents

Education

Bachelor of Technology in Computer Science and Engineering
Vellore Institute of Technology, Amaravati, India Expected: June 2026
GPA: 8.20*(7th sem)/ 10.0
Thesis: *[Context Aware Decision Making agent for Smart Assistants]* Advisor: Dr. Rajkumar Yesuraj

Research Experience

Undergraduate Research Assistant — AIR Center May 2024 – December 2025
VIT-AP University, Amaravati, India Advisor: Dr. Rajkumar Yesuraj

- Implemented AR-based simulation frameworks in Unity + ROS + OpenCV to support experiments on context-aware adaptive interfaces.
- Developed data pipelines for multimodal input (speech, gesture, visual signals) and performed statistical analysis using Python/SPSS.
- Integrated prototype adaptive decision-making models with reinforcement learning and rule-based policies for proof-of-concept testing.

Project Assistant — IIEC Center Aug 2023 – Jul 2024
VIT-AP University, Amaravati, India Advisor: Dr. Guruprakash Jayabalasamy

- Investigated adaptive user interface models in AR-supported human-robot collaboration environments.
- Conducted a structured literature review for a survey on vision-language interaction models in HCI.
- Collaborated on experiment design for evaluating multimodal adaptive systems.

Publications

1. Ray, T., Yesuraj, R. **A Comprehensive Survey on Human Behavior Prediction by Speech, Emotion and Gesture using Deep Learning Approaches**. Submitted to *[Journal of Ambient Intelligence and Humanized Computing]*, 2025. Under Review
2. Ray, T., Yesuraj, R. **Segment-Based Dynamic Programming for Optimal Binary Search Trees: A Sub-Cubic Algorithm with Hierarchical Weight Partitioning**. Submitted to *[Springer Algorithmica]*, 2026. Under Review
3. Ray, T., Yesuraj, R. **Context-Aware Emotionally Adaptive Voice Assistants: A Multi-Modal Framework for Empathetic Human-Agent Interaction**. Submitted to *[Journal of Automation and Intelligence]*, 2025. Under Review
4. Ray, T., Yesuraj, R. **Interactive Ethical Constraint Specification for LLM Decision Support: A Human-Centered Algorithmic Framework**. Submitted to *[Autonomous Agents and Multi-Agent Systems]*, 2025. Under Review
5. Ray, T., Yesuraj, R. **Synchronous Emotional Dynamics in Human-AI Collaborative Networks: A Temporal Graph Neural Network Approach**. Presented in Conference *[IEEE CICN]*, 2025
6. Ray, T., Rajkumar, Y., Shalini, R., Srigayathri, K., Jayashree, S. **XMACNet: An Explainable Lightweight Attention based CNN with Multi-Modal Fusion for Chili Disease Classification**. Published in Conference *[Springer's LNICST]*, 2025.
7. Ray, T., Rajkumar, Y. **Segment Based Dynamic Programming For Optimal Binary Search Trees: A Sub-Cubic Algorithm With Hierarchical Weight Partitioning**. Presented in *[National Conference on Undergraduate Research (NCUR) 2026- Abstract Submission ID: 27596]*, 2026.

Projects

Context Aware Decision Making agent for Smart Assistants — Capstone Project(Ongoing)
Deploying through multimodal signal fusion (voice, wearables) with on-device processing and RL-based adaptive decision policies, ensuring real-time, privacy-preserving, and user-centric smart assistant interactions.

AR-Enhanced Human-Robot Collaboration Interface
Multimodal interface design using Unity and ROS to support spatial collaboration and task switching with context-aware feedback.

Machine Vision based Third Eye for Safe Driving — Engineering Clinics
Developed a non-invasive eye-tracking interaction model using WebGazer.js and OpenCV to enable navigation in constrained environments.

Awards and Recognition

- Selected as **Student Mentoring Workshop (SMeW)**, [ICSE Conference 2026] – Rio de Janeiro, Brazil — 2025
- **ACM SIGPLAN PAC** Award for [POPL Conference, 2026] – Rennes, France — 2025
- **Finalist**, [National level MARK-XXIV Hackathon] – Woxsen University, India — 2023
- **Merit-based** Academic Excellence Scholarship (International Scholarship for Education & Enrichment (I-SEE) – VIT University, India — 2022

Technical Skills

- **Languages:** Python, Java, R, JavaScript, MATLAB, SQL
- **Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, StyleGAN
- **Tools:** NumPy, Pandas, SciPy, SPSS, RStudio, Unity, OpenCV, ROS
- **Design:** Figma, Adobe XD, Matplotlib, Seaborn, Tableau
- **Platforms:** Git, LaTeX, Overleaf, SPSS, Jupyter, Mendeley/Zotero

Volunteering

- **Student Volunteering(SV):** Principles of Programming Languages Conference **(POPL 2026)**-Rennes, France
- **Rotaract Club of VIT-AP UNIVERSITY:** Food donation, helping the underprivileged, organizing health camps, environmental clean-ups, educational support.